HTTP 5221

Security & Usability

Sean Doyle

INTRO TO USABILITY / HEURISTIC EVALUATIONS

Lesson 1

Sean Doyle (with materials from Simon Borer, Jemi Choi, and Andrei Navumau)

Course overview

Usability (includes accessibility).

Usability is **how useful something is**.

- **who's going to use it**,
- get data about **how well it works for them**,
- **interpret that data**
- **make recommendations**

Why does it matter?

In most of your courses, you learn **how to make things**.

In these modules, you will learn how to make things **for real people, better.**

Module 1 concepts:

1. Define general terms and concepts of both usability and heuristic evaluations.
2. Learn how to perform a heuristic evaluation.

Why Usability and Accessibility Matter

- Creates inclusive and engaging experiences.
- Ensures that products and systems are easy to use, efficient, and enjoyable for users.
- access to digital content and services regardless of abilities or disabilities

The Growing Significance of Usability & Accessibility

- Essential for user satisfaction and business success.
- Can improve customer loyalty, increase user engagement, and drive positive brand perception.

What is Usability?

- Ease of use and effectiveness of a product or system
- how well users can navigate the interface to achieve their goals

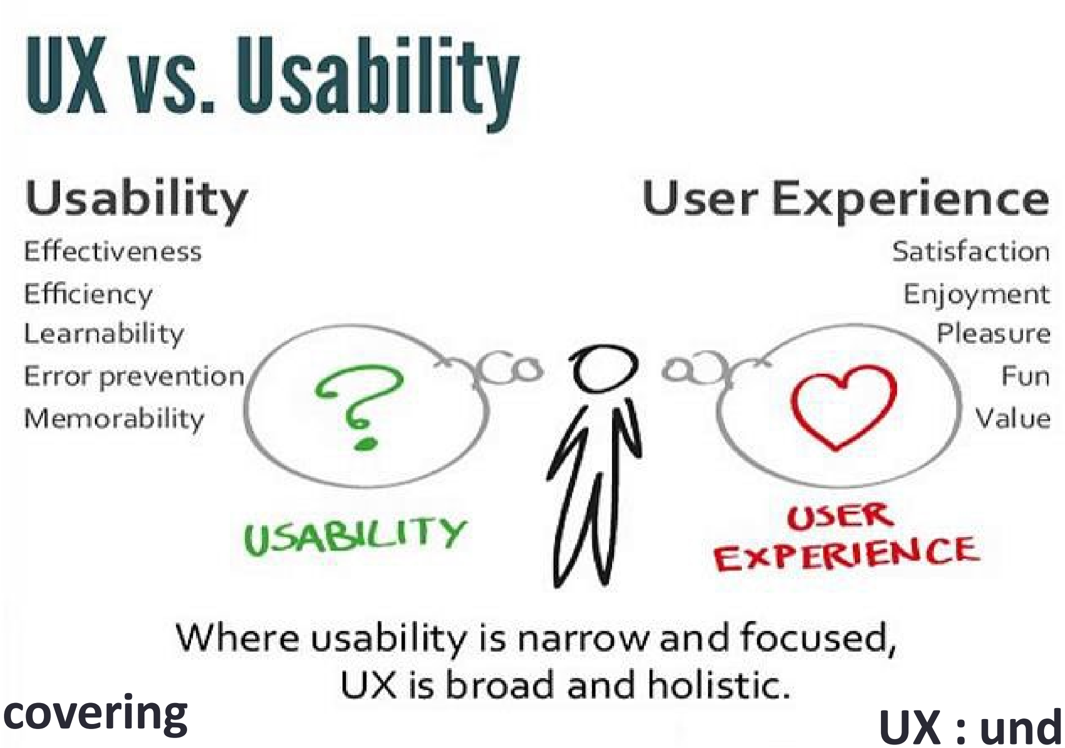
USABILITY

The extent to which specified users can find, understand and use information and services online. Web usability can be measured through the effectiveness and efficiency with which users can complete defined tasks online.

Government of Canada, Standard on Web Usability

Things that are *not* usability: UX

User experience (UX) is how people **feel** about their experience of using the product

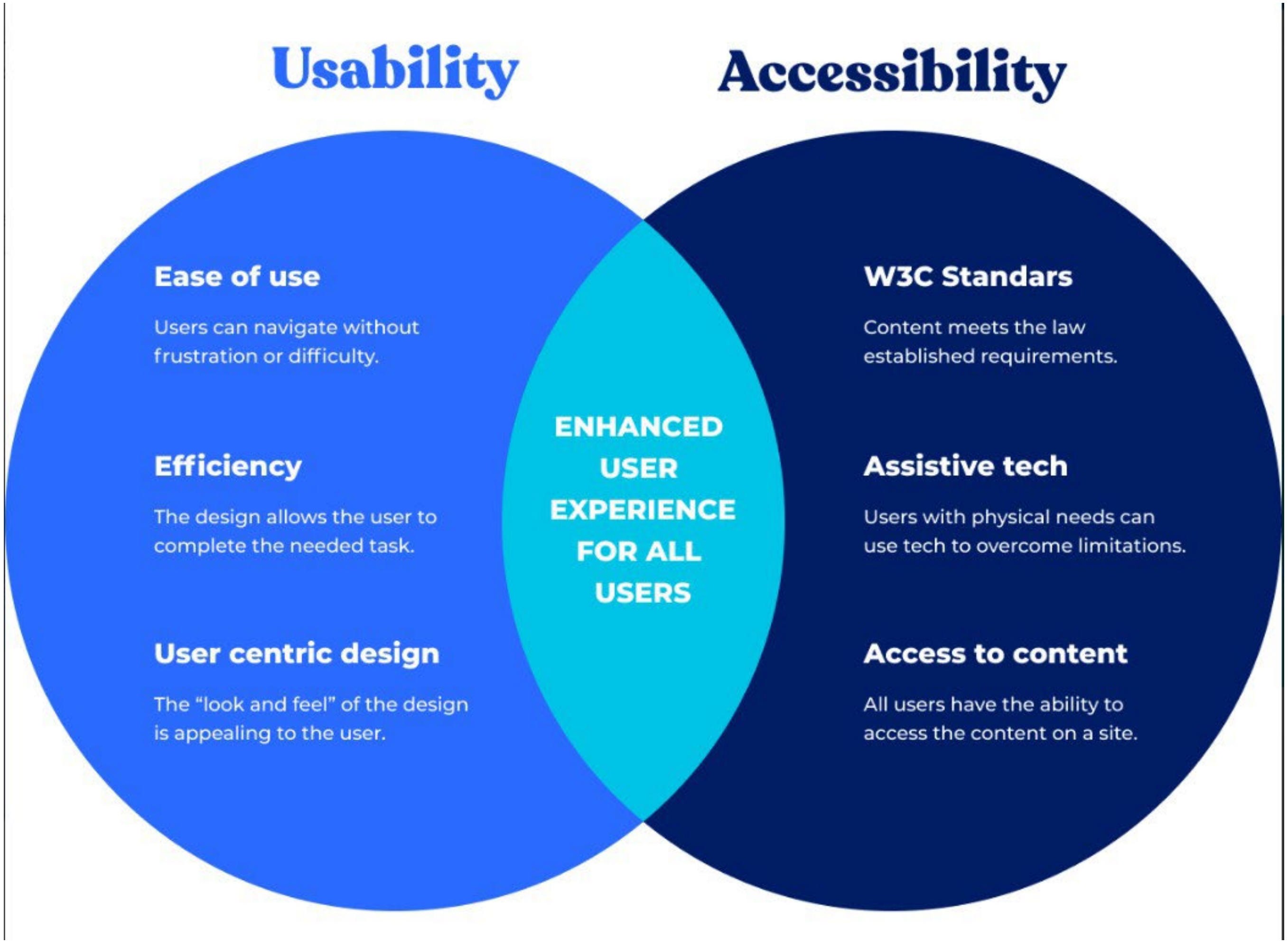


Usability : uncovering problems, stress points

UX : understanding positive experiences

Things that are *not* usability: Accessibility

Accessibility is how people with impairments (temporary, permanent or situational) can access information



Source: https://blankfactor.com/insights/blog/disabilities-and-ux-accessibility/

Usability === average **number of steps** to listen to a song.

UX === people ***like*** using the ipod

Accessibility === Can someone who has low vision or is blind find and play a song?

Accessibility

“The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect.”

“The Web is fundamentally designed to work for all people, whatever their hardware, software, language, location, or ability. When the web meets this goal, it is accessible to people with a diverse range of hearing, movement, sight, and cognitive ability.”

Tim Berners-Lee, W3C Founding Director and inventor of the World Wide Web

Source: https://www.w3.org/mission/accessibility/

Heuristics

“A heuristic is a mental shortcut that allows an individual to make a decision, pass judgment, or solve a problem quickly and with minimal mental effort.

While heuristics can reduce the burden of decision-making and free up limited cognitive resources, they can also be costly when they lead individuals to miss critical information or act on unjust biases.”

-https://www.psychologytoday.com/ca/basics/heuristics

Using a checklist of standardized principles.

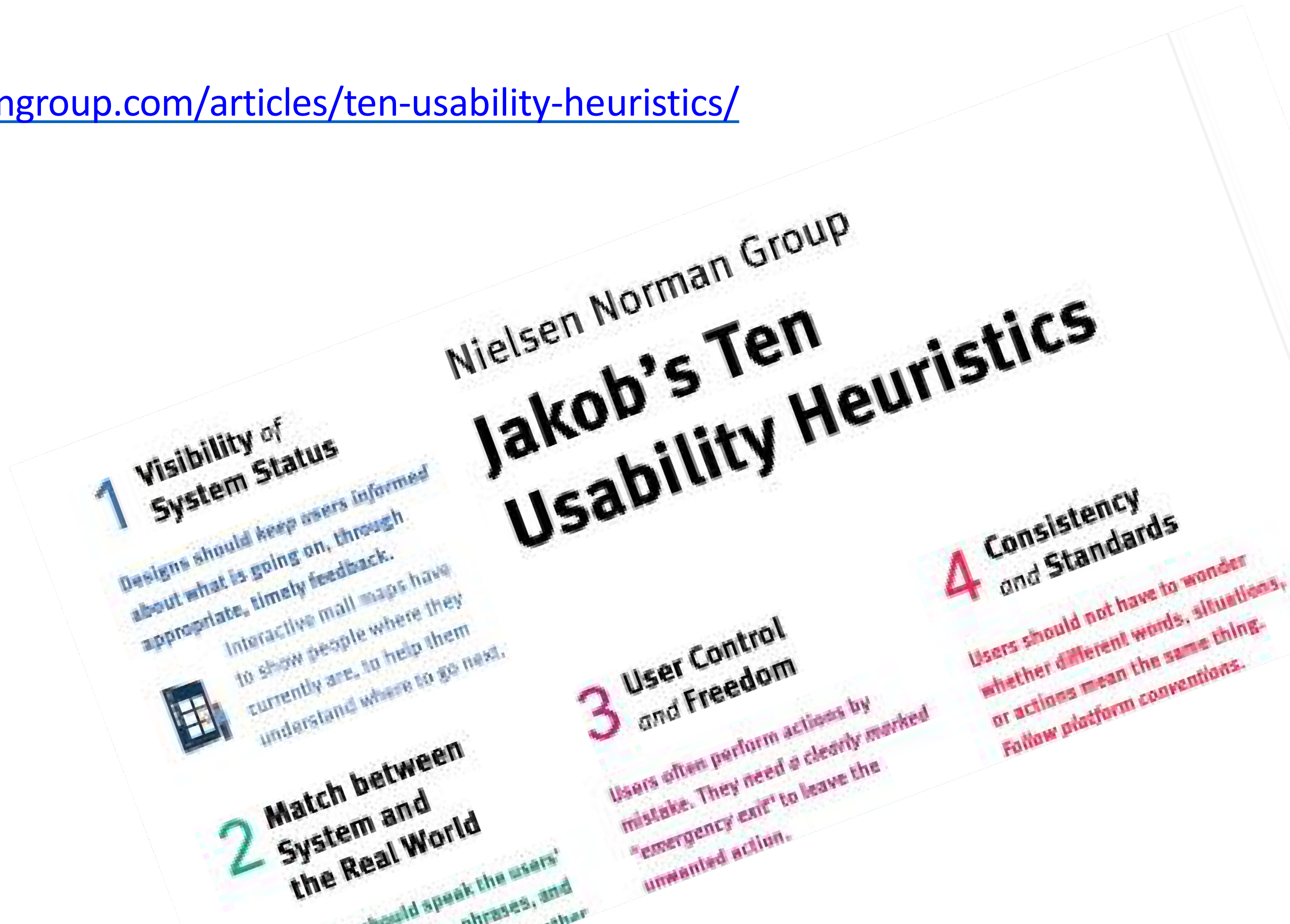
Heuristic Evaluations

- A heuristic evaluation is a type of usability auditing based on a **checklist of good ideas**.
- A group of 'experts' generally yields better results than auditing separately.

Jakob's Heuristic

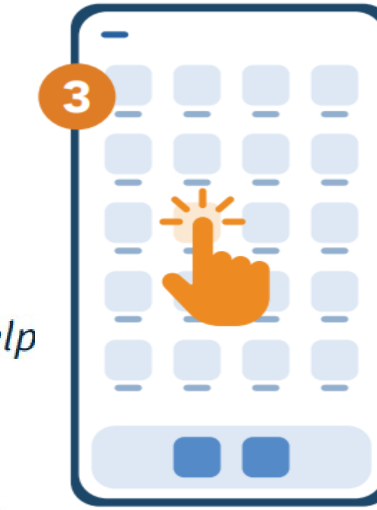
10 Usability Heuristics for User Interface Design

https://www.nngroup.com/articles/ten-usability-heuristics/



Jakob's Heuristic

#1: Visibility of System Status



1 ***"You Are Here" maps***

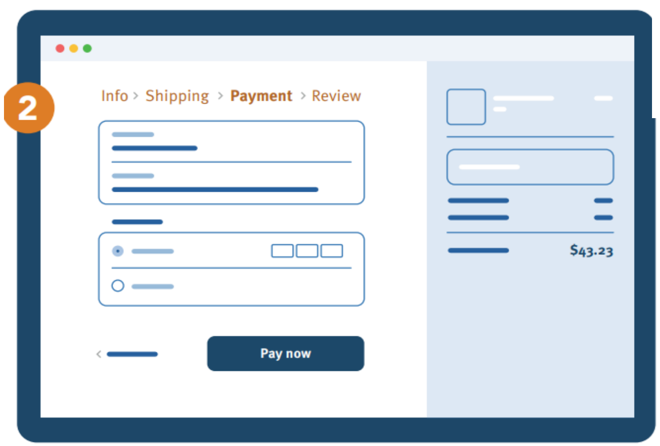
Interactive mall maps have to show people where they currently are, to help them understand where to go next.

2 ***Checkout flow***

Multistep processes show users which steps they've completed, they're currently working on, and what comes next.

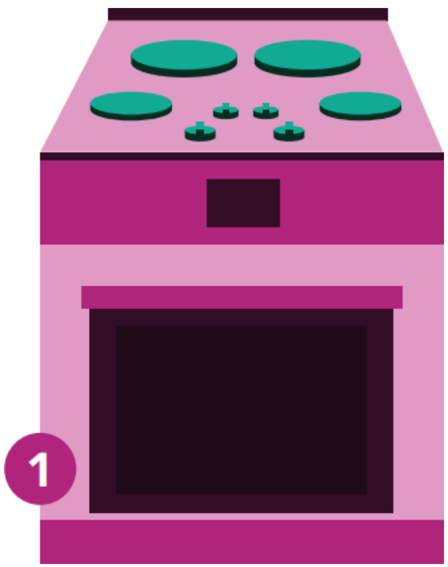
3 ***Phone tap***

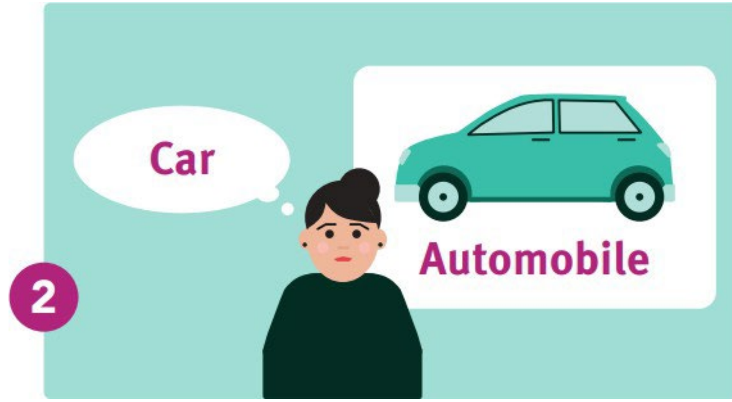
Touchscreen UIs need to reassure users that their taps have an effect — often through visual change or haptic feedback.



Jakob's Heuristic

#2: Match between system and the real world





1 *Stovetop controls*

When stovetop controls match the layout of heating elements, users can quickly understand which control maps to each heating element.

2 *"Car" vs. "automobile"*

If users think about this object as a "car," use that as the label instead.

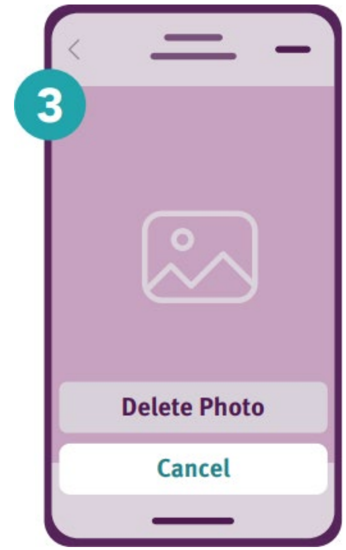
3 *Shopping cart icon*

A shopping cart icon is easily recognizable because that feature serves the same purpose as its real-life counterpart.

Jakob's Heuristic

#3: User control and freedom

Users often perform actions by mistake. They need a clearly marked **"emergency exit"** to leave the unwanted action without having to go through an extended process.



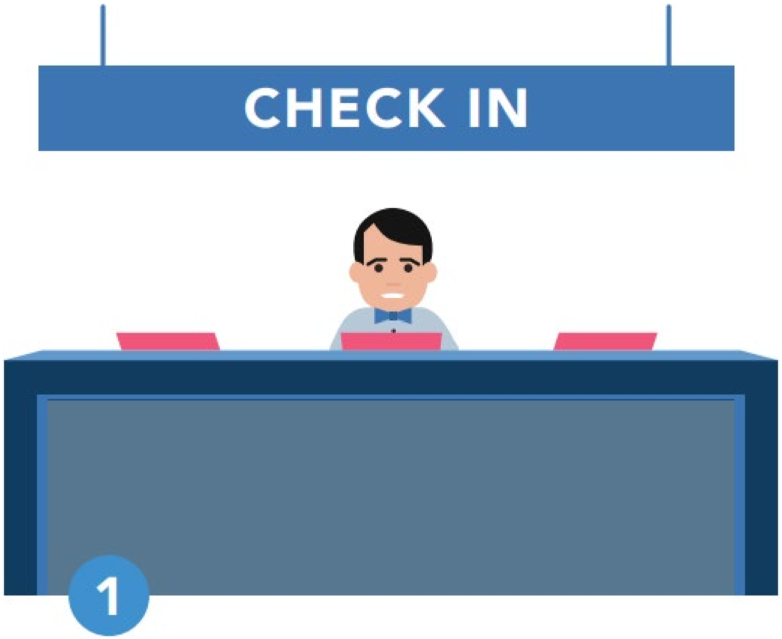
1. ***Exit sign***
   Digital spaces need quick "emergency" exits, just like physical spaces do.

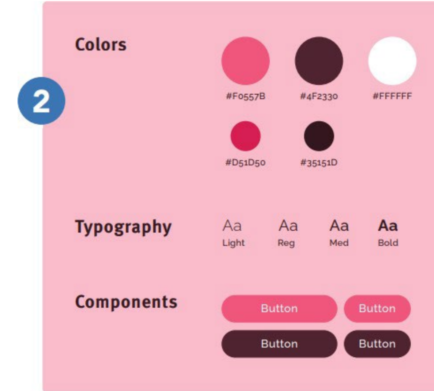
2. ***Undo and redo***
   These functions give users freedom because they don't have worry about their actions — everything is easily reversible.

3. ***Cancel button***
   Users shouldn't have to commit to a process once it's started — they should be able to easily cancel and abandon.

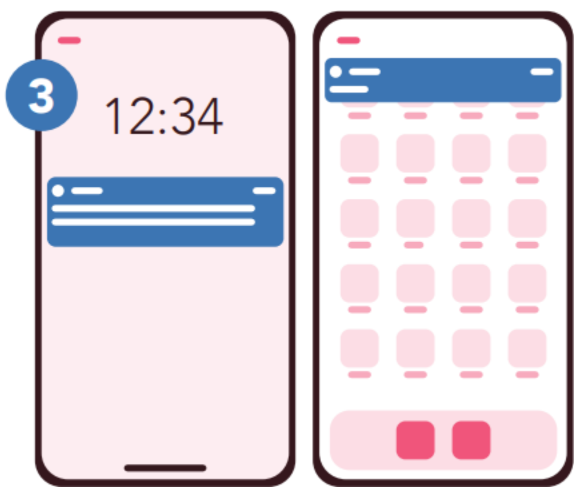
Jakob's Heuristic

#4: Consistency and standards





Users should not have to wonder whether different words, situations, or actions mean the same thing. **Follow platform and industry conventions.**



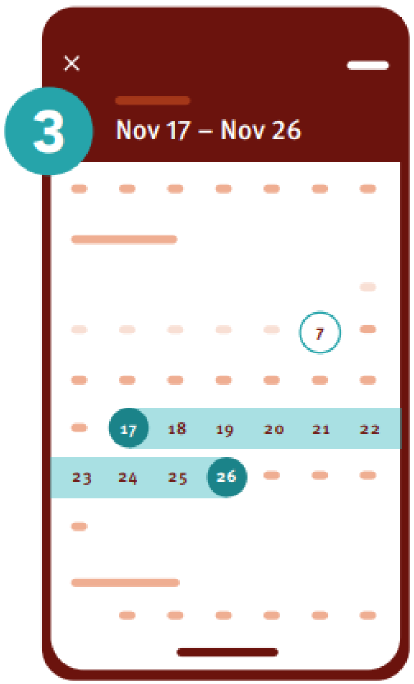
1. ***Check-in counter***
   Check-in counters are usually located at the front of hotels. This consistency meets customers' expectations.

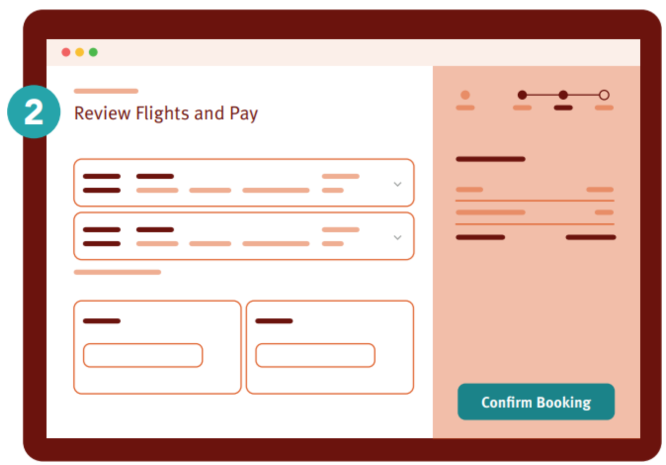
2. ***Design system***
   Using elements from the same design system across the product lines lowers the learning curve of users.

3. ***Notifications***
   A standardized notification design provides a similar but distinguishable look and feel for different app pop-ups.

#5: Error prevention

Good error messages are important, but the best designs **carefully prevent problems** from occurring in the first place.





1. ***Guard rails***
   Guard rails on curvy mountain roads prevent drivers from falling off of cliffs.
2. ***Airline confirmation***
   The confirmation page before checking out on airline websites gives users another chance to review the flight details.
3. ***Date selection on calendar***
   Offer good defaults and set boundaries when people book services by dates. Grey out unavailable options.

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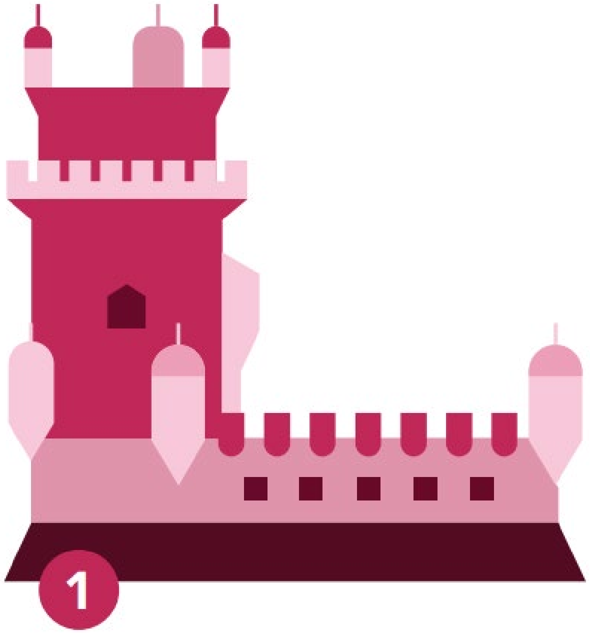
#6: Recognition rather than recall



Minimize the user's memory load by making elements, actions, and options visible. The user should **not have to remember information** from one part of the interface to another. Information required to use the design should be visible or easily retrievable when needed.

1 ***Lisbon***

People are more likely to correctly answer the question "Is Lisbon the capital of Portugal?" rather than "What's the capital of Portugal?"

2 ***Comparison table***

Comparison tables list key differences so that users don't need to remember them to make comparisons.

3 ***Search***

Search queries are presented together with the results as a reference.

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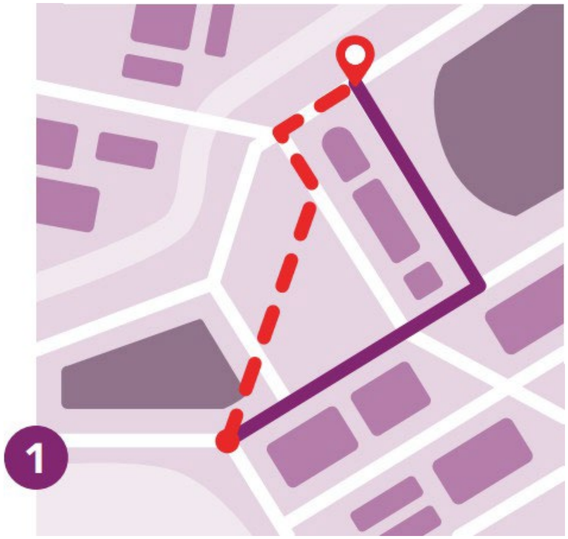
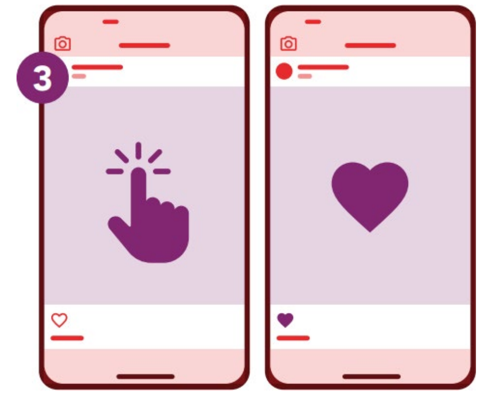
#7: Flexibility and efficiency of use



Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can **cater to both inexperienced and experienced users.** Allow users to tailor frequent actions.



1 ***Shortcuts***

Regular routes are listed on maps, but locals with more knowledge of the area can take shortcuts.

2 ***Keyboard shortcut***

Keyboard shortcuts for complex products can help expert users finish their tasks more efficiently.

3 ***Tap to like***

Social apps allow two ways to like posts. Experienced users can tap to like because it speeds up their browsing.

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#8: Aesthetic and minimalist design

Interfaces should not contain information which is irrelevant or rarely needed. Every extra unit of information in an interface **competes** with the relevant units of information and diminishes their relative visibility.

1 ***Ornate vs. simple teapot***

Excessive decorative elements can interfere with usability.

2 ***Communicate, don't decorate***

Over-decoration can cause distraction and make it harder for people to get the core information they need.

3 ***Messy vs organized UI***

Messy UI increases the interaction cost for users to find their desired content; Organized UI lowers the cost.

One of our favorite slogans

Jakob's Heuristic

#9 Help users recognize, diagnose, and recover from errors

Error messages should be expressed in **plain language**, specify the problem, and give constructive solution.

1 ***Wrong way sign***

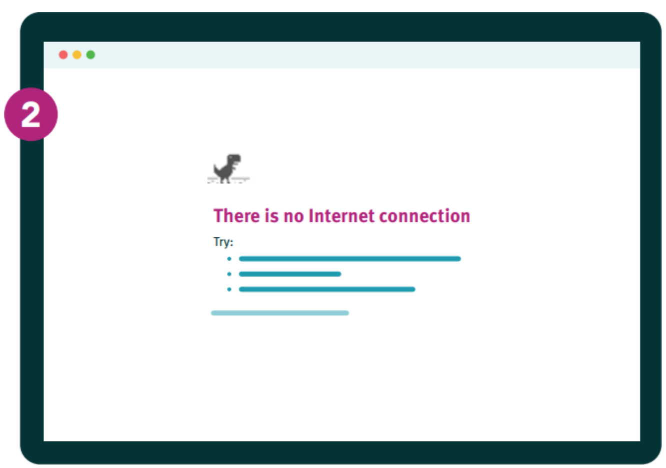
Wrong-way signs on the road remind drivers that they are heading in the wrong direction and ask them to stop.

2 ***Internet connection error***

Good internet connection error pages show what happened and constructivel instruct users on how to fix the problem

3 ***No search results***

Provide useful help when people encounter search-result pages returnin zero results, such as popular topics.



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#10: Help and documentation

It's best if the system **doesn't need** any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

1 *Airport information center*

Information kiosks at airports are easily recognizable and solve customers' problems in context and immediately.

2 *Frequently asked questions*

Good frequently-asked-question pages anticipate and provide the helpful information that users might need.

3 *Information icon*

Information icons reveal tooltips to explain jargon when users touch or hover over them, which provides contextual help.

HEURISTIC EVALUATION

Conducting Heuristic Evaluation Using Jakob Nielson’s 10 Usability Heuristics

https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/

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